Functional Equations — problem 11.25

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Does there exist a function $f : \mathbb{N} \mapsto \mathbb{N}$ such that

$$f(1) = 2, \quad f[f(n)] = f(n) + n, \quad f(n) < f(n+1) \quad \text{for all } n \in \mathbb{N} \qquad \text{(IMO 1993)}\,?$$